Learning Journey — M.Tech Data Science & Engineering (BITS Pilani WILP, 2024–26)

What I studied each semester, what I delivered, and why it matters for the work I do. The dissertation is the capstone that pulls these subjects together; the rest of the programme is the foundation under it.

> **How to read this.** Each subject has the same four lines: **Covered** (what the course taught), **Delivered** (what I built or submitted), **Skills** (what I can now do) and **Value** (how it adds to my experience). Group assignments are described at a high level only; the source files are not published here (see Coursework/README).

Programme at a glance

Year / Semester	Subjects
Year 1 · Semester 1 (2024–25)	Mathematical Foundations for Data Science · Data Structures & Algorithms Design · Computer Organization & Software Systems · Introduction to Data Science · *(audit)* Python Fundamentals for Data Science
Year 1 · Semester 2 (2024–25)	Data Visualization & Interpretation · Introduction to Statistical Methods · Applied Machine Learning · Data Management for Machine Learning
Year 2 · Semester 1 (2025–26)	Stream Processing & Analytics · Big Data Systems · Information Retrieval · Natural Language Processing
Year 2 · Semester 2 (2025–26)	Dissertation: *A Hybrid Scoring Framework for Jira Requirement Quality*

How it builds up: Year 1 lays the foundations (maths, algorithms, systems, the data-science process, statistics, ML, data management). Year 2 moves to scale and language (streaming, big data, retrieval, NLP), and the dissertation pulls it all into one working system.

Themes that run across semesters

Theme	Subjects	Where it shows up in the capstone
Maths and statistics	Mathematical Foundations · Statistical Methods	TF-IDF/linear models, evaluation metrics, significance of results
Programming and engineering	Python · Data Structures & Algorithms · Computer Organization	Clean, testable Python package (src/, tests/)
Data work end to end	Intro to Data Science · Data Management for ML · Big Data · Stream Processing	Data preparation, repeatable pipelines, reasoning about scale
Machine learning and language	Applied ML · NLP · Information Retrieval	TF-IDF text classifier, class balancing, model comparison
Communicating insight	Data Visualization	Streamlit dashboard, explainable scores

The through-line: I can take a messy, real-world problem, frame it, prepare the data, choose and evaluate a model, and put the result in front of the people who must act on it. That combination of data engineering, ML and communication is the core of an AI product role.

Year 1 · Semester 1 (2024–25)

Python Fundamentals for Data Science (audit course)

- **Covered:** Python basics, strings/lists/sets/tuples/dictionaries, control flow, functions, files, NumPy, pandas, Matplotlib and Seaborn.
- **Delivered:** six sessions of worked notebooks plus small exercises (factorial, list intersection, duplicate removal, character count) in `Coursework/python/`.
- **Skills:** fluent data wrangling and plotting in Python.
- **Value:** the base layer for everything else, including the dissertation's data pipeline.

Mathematical Foundations for Data Science (DSECL ZC416)

- **Covered:** linear algebra (systems, vector spaces, eigen/SVD decompositions), vector calculus and gradients, backpropagation, optimisation (gradient descent, constrained and convex), PCA, SVM optimisation.
- **Delivered:** assignments covering problems across the learning areas above.
- **Skills:** reading the maths behind ML papers and library docs; knowing *why* gradient descent, regularisation and PCA behave as they do.
- **Value:** lets me question model behaviour instead of treating algorithms as black boxes, which is useful when judging vendor or team claims.

Data Structures and Algorithms Design (DSECL ZG519)

- **Covered:** algorithm analysis, trees, heaps, graphs, bloom filters, sorting/searching, greedy, divide-and-conquer, dynamic programming, MapReduce, NP-completeness.
- **Delivered:** a team design-and-code assignment (problem set 07) using **breadth-first search on a river-network graph** to place flood-control dams, plus an employee-eligibility problem implemented in Python, each with a design document.
- **Skills:** choosing the right data structure, reasoning about complexity, modelling real problems as graphs.
- **Value:** better engineering judgement when working with developers on feasibility, performance and trade-offs.

Computer Organization and Software Systems (DSECL ZG516)

- **Covered:** processor and memory organisation, caches, I/O, operating-system role (processes, threads, virtual memory), performance measures such as MIPS and Amdahl's law.
- **Delivered:** two simulator-based assignments using the CPU-OS Simulator.
    1. **Cache behaviour:** Part I direct mapping, Part II associative mapping, Part III set-associative mapping.
    2. **Pipelining, scheduling and multithreading:** Part I pipeline architecture, Part II scheduling algorithms (FCFS, SJF, Round Robin), Part III multithreading.
- **Skills:** relating software performance to hardware and OS behaviour.
- **Value:** helps me reason about latency, throughput and resource cost when scoping data and AI workloads.

Introduction to Data Science (DSECL ZG532)

- **Covered:** CRISP-DM and the data-science process, data quality, preprocessing and feature engineering, classification, association rules, clustering, anomaly detection, storytelling with data.
- **Delivered:** a team assignment analysing **Spotify vs YouTube song popularity** (business framing, data acquisition, cleaning, EDA, feature engineering, classification and clustering). My own full solution notebook is in `Coursework/data-science/`.

- **Skills:** running a data project through every stage, with decisions documented.
- **Value:** the same structure I follow when scoping any data or ML initiative.

Year 1 · Semester 2 (2024–25)

Data Visualization and Interpretation (DSECL ZG555)

- **Covered:** visual perception, choosing the right chart, storytelling, dashboard design principles, Tableau, Matplotlib, Seaborn and Bokeh.
- **Delivered:** two team assignments. (1) A **Tableau dashboard** on India's petroleum and natural-gas statistics, designed for a leadership audience. (2) An **employee-attrition analysis** identifying the features with the highest impact, in Python.
- **Skills:** designing for the audience first, then the chart; building dashboards that answer a decision.
- **Value:** I can present analytics so stakeholders act on it. The dissertation's dashboard applies the same principles.

Introduction to Statistical Methods (DSECL ZC418)

- **Covered:** probability, Bayes' theorem, distributions, estimation, hypothesis testing, maximum likelihood, ANOVA, time series (AR/MA/ARIMA/SARIMA), Gaussian mixtures and EM.
- **Delivered:** assignments covering problems across the learning areas above.
- **Skills:** designing sound comparisons, interpreting significance and confidence, basic forecasting.
- **Value:** supports credible A/B tests, metric reviews and honest reporting of model performance (for example reporting variance across cross-validation folds in the dissertation).

Applied Machine Learning (DSECL ZG568)

- **Covered:** end-to-end ML projects, preprocessing, model selection, gradient descent, decision trees, perceptrons and multi-layer networks, CNNs, interpretability and limitations.
- **Delivered:** an individual **Streamify customer-churn prediction** assignment (EDA, preprocessing, Logistic Regression and Random Forest, evaluation, recommendations): `Coursework/applied-machine-learning/`. I also completed a linear-regression exercise (single and multiple variables): `Coursework/ml-regression-exercise-1/`.
- **Skills:** building and evaluating classifiers and turning results into business recommendations.
- **Value:** the model-building core I reused in the dissertation's classifier comparison.

Data Management for Machine Learning (DSECL ZG529)

- **Covered:** data models and query languages, ML workflow and its data challenges, data pipelines and their stages, modern data platforms (warehouses, lakes, orchestration), model metadata and registry, observability, privacy and anonymisation.
- **Delivered:** a customer-churn data-pipeline project using **DVC (data version control)**.
- **Skills:** versioning data and models, building repeatable pipelines, thinking about metadata, monitoring and privacy up front.
- **Value:** the "MLOps" side of product work: what must exist so an ML feature is reproducible, auditable and maintainable.

Year 2 · Semester 1 (2025–26)

Stream Processing and Analytics (DSECL ZG556)

- **Covered:** architecture of streaming systems, stream algorithms and data structures, tools for real-time analytics, streaming SQL and streaming ML.
- **Delivered:** two team assignments. (1) A **white paper on serverless streaming architectures**: formulating a real-time application scenario and mapping it to cloud patterns. (2) A **PySpark MLlib** workflow on 20 years of XAU/USD gold-price data (prepare, train, predict, interpret).
- **Skills:** deciding when batch is not enough, designing real-time pipelines, applying ML to time-stamped data.
- **Value:** relevant to any "real-time" product requirement, such as fraud and risk signals or live monitoring.

Big Data Systems (DSECL ZG522)

- **Covered:** distributed storage and processing, Hadoop ecosystem, NoSQL databases, Spark (SQL, streaming, ML), cloud storage and database services.
- **Delivered:** two team assignments. (1) **MongoDB:** loading a road-accident dataset with `mongoimport` and answering analytical questions with aggregation queries. (2) **PySpark:** analysing a large Amazon product-review dataset (loading, schema fixes, cleansing, Spark SQL analytics), with a written report and demo video.
- **Skills:** choosing between relational, document and distributed engines; writing Spark DataFrame/SQL jobs.
- **Value:** I can discuss scale, cost and architecture trade-offs with data engineers credibly.

Information Retrieval (DSECL ZG537)

- **Covered:** Boolean and vector-space retrieval, tolerant retrieval, index construction and compression, scoring and term weighting, text classification and clustering, evaluation, web search, recommender systems, neural IR.
- **Delivered:** two team assignments on one problem set. (1) An **inverted index over legal case reports** with **spelling correction**, comparing standard and weighted edit distance. (2) **Index construction and compression** for a digital library (tokenisation, normalisation, stemming, compression).
- **Skills:** how search systems index, rank and evaluate text; TF-IDF intuition.
- **Value:** search and retrieval are central to AI assistants and RAG-style products; the dissertation's chatbot uses the same retrieval ideas.

Natural Language Processing (DSECL ZG530)

- **Covered:** language models and smoothing, word embeddings, neural LMs, part-of-speech tagging and HMMs, parsing, topic models.
- **Delivered:** two team assignments. (1) **N-gram language models** and text-similarity analysis on Twitter and Reddit data. (2) **LDA topic modelling** on Amazon Fine Food reviews (coherence scoring, topic visualisation, dependency parsing).
- **Skills:** text preprocessing, language-model evaluation, extracting themes from large text collections.
- **Value:** directly underpins the dissertation (text quality signals) and any feedback-mining or requirements-analysis feature.

Year 2 · Semester 2 (2025–26): Capstone

Dissertation — *A Hybrid Scoring Framework for Jira Requirement Quality* (DSECL ZG628T)

- **Covered:** applied research method: problem definition, literature review, experiment design, evaluation, written report and final presentation.
- **Delivered:** a working system (`Dissertation/`): a 10-rule weighted engine (sums to 100) combined with a TF-IDF classifier (Random Forest / Logistic Regression / Linear SVM, SMOTE-balanced) into a tunable hybrid score and quality gate; a Streamlit dashboard with per-ticket explanations and a conversational auditor; 12 automated tests; a 55-page report, slides and a demo video. Results: ~92 % hold-out accuracy and ~0.96 F1 in 5-fold cross-validation on 525 labeled tickets (synthetic dataset).
- **Skills:** framing a vague quality problem as a measurable one; hybrid rule + ML design; handling class imbalance; explainability; shipping an app people can actually use.
- **Value:** shows I can own a problem from idea to usable tool, and that I choose *explainable* approaches where users need to trust the output. This is the most direct evidence for product-minded AI work.

How the programme adds up

1. **Breadth with depth:** maths and statistics for rigour, engineering for building, data platforms for scale, ML and language for intelligence, visualisation for adoption.
2. **Evidence, not just coursework:** a working capstone with tests, a report and a demo, plus individual notebooks that show my own working.
3. **Product relevance:** I can translate between business need, data reality and model capability, which is the daily work of an AI product role. This repository sits alongside my AIPM portfolio, which covers the product-management side.